

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (EC) Examination

Dec 2012

EC406 Digital Image Processing (Elective - I)

Date: 08/12/2012, Saturday

Time: 10.00 am to 01.00 pm

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Define: 1) Picture elements 2) Digital Image 3) Mid level Processing [03]
- (b) Define distance measures between pixels of an image: [02]
- (i) Euclidean distance (ii) City-block distance.
- (c) What is histogram of an image? Explain the concept of histogram equalization. [02]
- Q - 2 (a) Describe the electromagnetic spectrum based different fields of digital image processing. [07]
- (b) For the two image subsets, S_1 and S_2 as shown in below figure, for $V = \{1\}$ determine [07]
- whether these two subsets are (i) 4-adjacent (ii) 8-adjacent or (iii) m-adjacent.

S_1					S_2				
0	0	0	0	0	0	0	1	1	0
1	0	0	1	0	0	1	0	0	1
1	0	0	1	0	1	1	0	0	0
0	0	1	1	1	0	0	0	0	0
0	0	1	1	1	0	0	1	1	1

OR

- Q - 2 (a) Explain basic concepts of image sampling and quantization process. [07]
- (b) Two images, $f(x,y)$ and $g(x,y)$, have histograms h_f and h_g . Give the conditions under which you can determine the histograms of [07]
- (i) $f(x,y) + g(x,y)$ (iii) $f(x,y) - g(x,y)$
- (ii) $f(x,y) \times g(x,y)$ (iv) $f(x,y) \div g(x,y)$
- Q - 3 (a) Describe basic of spatial filters? Compare sharpen filters and smoothing filters in spatial domain. [07]
- (b) Show that the Laplacian operation is isotropic means invariant to rotation. Use [07]
- following equations relating co-ordinates after axis rotation by an angle θ .

$$x = x' \cos \theta - y' \sin \theta$$

$$y = x' \sin \theta + y' \cos \theta$$

Where (x,y) are the unrotated and (x', y') are the rotated coordinates.

OR

- Q - 3 (a) What is gray level transformation? Explain each with application. [07]
 (b) What are the fundamental steps required in digital image processing? What are the components required in image processing system? [07]

SECTION - II

- Q - 4 (a) Explain basics of filtering in frequency domain. [03]
 (b) What is blind deconvolution? [02]
 (c) Describe pseudo color image processing. [02]
- Q - 5 (a) Suppose to form a low pass filter that averages the four immediate neighbors of a point (x, y) but excludes the point itself. [07]
 (i) Find the equivalent filter $H(u, v)$ in the frequency domain.
 (ii) Show that this result as a low pass filter
 (b) Explain in detail: Image Sharpening using frequency domain filters. [07]

OR

- Q - 5 (a) What is the Homomorphic filter? Describe significance of Homomorphic filter. [07]
 (b) Compute Mean filters (Arithmetic, Geometric and Harmonic) and Order statistics filters (Max, Min and Median) value (for each case) of center pixel of given 3 X 3 Image. Prepare a Matlab code or through mathematical based method for computation. [07]

1	7	5
6	2	3
1	4	2

- Q - 6 (a) Explain Adaptive Median Filtering Algorithm with example. [07]
 (b) What is RGB color model? What are the limitations of RGB model? Compare with the HSI color model. [07]

OR

- Q - 6 (a) Describe: Color segmentation of an image. [07]
 (b) Smoothing and Sharpening in Color image processing. [07]
